

ADVANCED GEOPOLITICAL STUDIES & WORLD GEOGRAPHY

New Egyptian Baccalaureate Framework (EG-Bac Track) - Grade 12

ACADEMIC YEAR: 2025 / 2026 | EG-BAC GRADE 12 (ADVANCED HUMANITIES & GEOPOLITICS)

CURRICULUM OVERVIEW & AUTHORIZED SCOPE

Advanced analytical syllabus focusing on geospatial intelligence (GIS/RS), mathematical compactness modeling, international maritime law, collective security, and multipolar world transitions.

CORE CURRICULUM HIGHLIGHTS:

- Advanced GIS and satellite remote sensing case applications in border monitoring and resource management
- Quantitative state power modeling: demographic-economic equations and territory morphology indexes
- In-depth international law review of maritime jurisdictions, deep-sea mining, and sovereign shelf delimitations
- Critical geopolitical analysis of multinational corporations, unipolar vs multipolar paradigms, and supply chain security

APPROVED CHAPTER MODULES:

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|--|-----------------------------|
| - Ch 1: Advanced Spatial Analytics & Geographic Information Systems | pp. 1-26 (12 core units) |
| - Ch 2: State Morphology, Demographics & Modern Geopolitical Power Index | pp. 27-82 (22 core units) |
| - Ch 3: Maritime Delimitation Law, Hydrocarbon Basins & Geopolitical Conflicts | pp. 83-138 (20 core units) |
| - Ch 4: Global Economic Integration & Collective Security Architectures | pp. 139-174 (16 core units) |
| - Ch 5: Globalization Dynamics, Global Hegemony & Future World Order | pp. 175-212 (14 core units) |

OFFICIAL PUBLICATION METADATA & VERIFICATION

Document: EG-Bac Advanced Geopolitics, Maritime Law & Global Order Guide (36p - 3.6 MB)
Full Ministerial Textbook: 212 Pages - 31.2 MB (Direct on MOE Portal)
Publisher: Egyptian Ministry of Education - National Baccalaureate Directorate
Official Portal: <https://moe.gov.eg/ar/elearning/egbac/geography/> - Code: EGBAC-GE0-G1
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MINISTERIAL PREFACE & STATUTORY FRAMEWORK

MINISTERIAL DECREE NO. 138 OF 2025

Pursuant to the decisions of the Supreme Council for Pre-University Education, this textbook is officially accredited as the national core curriculum for Senior Secondary and Baccalaureate qualifications for the 2025/2026 academic cycle.

1. Pedagogical Scope & National Objectives

Advanced analytical syllabus focusing on geospatial intelligence (GIS/RS), mathematical compactness modeling, international maritime law, collective security, and multipolar world transitions. Advanced GIS and satellite remote sensing case applications in border monitoring and resource management Quantitative state power modeling: demographic-economic equations and territory morphology indexes In-depth international law review of maritime jurisdictions, deep-sea mining, and sovereign shelf delimitations Critical geopolitical analysis of multinational corporations, unipolar vs multipolar paradigms, and supply chain security

2. Assessment Cognitive Taxonomy Specification

All ministerial evaluation instruments conform strictly to the following weight distribution:

40% Knowledge & Recall Definitions, laws & formulas	35% Understanding & Application Calculations, proofs & models	25% Higher Order Thinking (HOTS) Analysis, synthesis & non-routine
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3. Approved Scientific Calculators & Examination Instrumentation

- Approved Calculators: Non-programmable scientific calculators (Casio fx-991ARX, fx-991CW, or equivalent approved models).
- Prohibited Devices: Any device with symbolic CAS (Computer Algebra System), programmable memory, internet connectivity, or graphic display capabilities is strictly prohibited in examination centers.
- Formula Reference Sheets: The official ministerial formula sheet is printed directly in the examination booklet; separate personal scratch notes are forbidden.
- Drawing Instruments: Standard geometric compass, protractor, 30/60 and 45 degree set squares, and transparent rulers are mandatory.

4. Egyptian Knowledge Bank (EKB) & Digital Portal Synchronisation

Students and teachers can access interactive simulations, GeoGebra 3D interactive graphs, video lectures by Senior Curriculum Inspectors, and self-assessment question banks at the official Ministry of Education digital portal: <https://moe.gov.eg/ar/elearning-content/>. Scanning the QR code on the back cover links directly to this textbook's digital resource pack.

SYLLABUS BLUEPRINT & EXAMINATION WEIGHT MATRIX

Official Duration: 180 Minutes (3 Hours) | Total Paper Marks: 60 Marks

Chapter / Modular Title	Weeks	Marks	Weight	Curricular Focus
Ch 1: Advanced Spatial Analytics & Geographic Information Systems	6 wks	12 pts	20.0%	12 core instructional units covering national standards and
Ch 2: State Morphology, Demographics & Modern Geopolitical Power Index	6 wks	12 pts	20.0%	22 core instructional units covering national standards and
Ch 3: Maritime Delimitation Law, Hydrocarbon Basins & Geopolitical Conflicts	6 wks	12 pts	20.0%	20 core instructional units covering national standards and
Ch 4: Global Economic Integration & Collective Security Architectures	6 wks	12 pts	20.0%	16 core instructional units covering national standards and
Ch 5: Globalization Dynamics, Global Hegemony & Future World Order	6 wks	12 pts	20.0%	14 core instructional units covering national standards and

INTERDISCIPLINARY COGNITIVE BRIDGES

Physics (Mechanics & Electromagnetism) Direct mathematical articulation with rotational kinetics, projectile kinematics, electric flux integrations, and harmonic oscillators.
Engineering Sciences (Civil, Mechanical, Electrical) Matrix state-space modeling, 3D spatial stress analysis, moments of inertia, and equilibrium structural design.
Computer Science & Algorithmics Discrete combinatorics, graph search trees, Boolean logic transformations, algorithmic asymptotic complexity, and linear transformations.

CHAPTER SPECIFICATION: CH 1: ADVANCED SPATIAL ANALYTICS & GEOGRAPHIC INFORMATION SYSTEMS

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Ch 1: Advanced Spatial Analytics & Geographic Information Systems.
- [+] Synthesize and apply analytical frameworks across all 12 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 1.1: Primary foundational doctrines and axiomatic structures of Ch 1: Advanced Spatial Analytics & Geographic Information Systems.

Curricular Analytical Directive 1.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_1 = \frac{\sum (\omega_k \cdot \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha \cdot f(x) + \beta \cdot g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: CH 2: STATE MORPHOLOGY, DEMOGRAPHICS & MODERN GEOPOLITICAL POWER INDEX

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Ch 2: State Morphology, Demographics & Modern Geopolitical Power Index.
- [+] Synthesize and apply analytical frameworks across all 22 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 2.1: Primary foundational doctrines and axiomatic structures of Ch 2: State Morphology, Demographics & Modern Geopolitical Power Index.

Curricular Analytical Directive 2.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_2 = \frac{\sum (\omega_k \cdot \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha \cdot f(x) + \beta \cdot g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: CH 3: MARITIME DELIMITATION LAW, HYDROCARBON BASINS & GEOPOLITICAL CONFLICTS

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Ch 3: Maritime Delimitation Law, Hydrocarbon Basins & Geopolitical Conflicts.
- [+] Synthesize and apply analytical frameworks across all 20 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 3.1: Primary foundational doctrines and axiomatic structures of Ch 3: Maritime Delimitation Law, Hydrocarbon Basins & Geopolitical Conflicts.

Curricular Analytical Directive 3.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_3 = \frac{\sum (\omega_k \cdot \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha \cdot f(x) + \beta \cdot g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

OFFICIAL MINISTRY FORMULA REFERENCE SHEET

AUTHORIZED CONCEPT CARD (SHEET OF CONCEPTS) PROVIDED IN EXAMINATION CENTERS

ADVANCED GEOPOLITICAL STUDIES & WORLD GEOGRAPHY -

Fundamental Identity: $\Psi(x) = \sum_{k=1}^n c_k \cdot \phi_k(x)$

Equilibrium Balance Criterion: $\sum F_{\text{net}} = 0$, $\sum M_{\text{pivot}} = 0$

Continuity & Conservation Standard: $\text{div}(\mathbf{J}) + \text{partial } \rho / \text{partial } t = 0$

Normalized Standard Score: $Z = (X - \mu) / \sigma$

PROCEDURAL ASSESSMENT & TRANSFORMATION RULES

Linear Operator Transformation: $T(\alpha * u + \beta * v) = \alpha T(u) + \beta T(v)$

Harmonic Synthesis Mean: $H = n / (\sum_{i=1}^n 1/x_i)$

Sample Variance Boundary: $s^2 = (1 / (n - 1)) * \sum (x_i - \bar{x})^2$

Curricular Efficiency Ratio: $\eta = (\text{Output_actual} / \text{Input_stand})$

NATIONAL EXAMINATION PROTOCOLS & EVALUATION MATRIX

1. Structure of the National Examination Paper

Paper Duration:	180 Minutes (3 Hours)
Total Scaled Marks:	60 Marks
Section I (Objective):	Multiple Choice Questions (MCQ) with 4 options; single best answer.
Section II (Subjective):	Structured multi-step analytical and proof essay questions.
Electronic Scoring:	Objective answer sheets processed via computerized optical mark recognition (OMR).
Manual Essay Scoring:	Double-blind evaluation by two independent certified ministerial examiners.

2. Standard Candidate Exam Directives

- All examination responses must strictly adhere to official ministerial syllabus specifications.
- Clear intermediate analytical steps, definitions, and proper notation are required for full credit.
- Calculators and reference sheets must strictly comply with official Ministry regulations.
- Review the certified digital textbook compendium for complete model solutions and rubrics.

ARAB REPUBLIC OF EGYPT - MINISTRY OF EDUCATION
NATIONAL CENTER FOR EXAMINATIONS & EDUCATIONAL EVALUATION (NCEEE)

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